

MS-DIAL Console Untargeted Plasma Lipidomics

Complete POSITIVE and NEGATIVE Mode Workflow

Thermo Orbitrap Exploris 240 • Human Plasma • DDA • mzML MS1–MS2
MS-DIAL Console 5.5.251021 • Linux/Ubuntu • MSP-based lipid MS/MS annotation

PURPOSE Provide a reproducible record of the MS-DIAL Console workflow actually executed for positive- and negative-ion plasma lipidomics, including the reason for each QC/checking step, the exact commands used, the observed outputs, and the files that should be carried forward. This document intentionally excludes LipidSearch, LIPID MAPS, LIPEA, and the previously discussed sample-specific suspicious-lipid review.

Parameter	Setting
Sample type	Human plasma
Instrument	Thermo Orbitrap Exploris 240
Acquisition	Data-dependent acquisition (DDA)
Ionization	Positive and Negative
Mobile phase additive	10 mM ammonium formate + 0.1% formic acid
Input format	mzML
MS levels	MS1–MS2
Processing software	MS-DIAL Console
Annotation library	MSP lipid spectral library
Operating environment	Ubuntu/WSL

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1. Scope

This handover covers two separate MS-DIAL Console LC-MS lipidomics analyses: one positive-ion run and one negative-ion run. They are processed independently and are not mixed during feature detection, annotation, or QC.

- Sample type: human plasma.
- Instrument: Thermo Orbitrap Exploris 240.
- Acquisition: DDA.
- Input: centroided mzML containing MS1 and MS2 data.
- Ionization: ESI positive or ESI negative, processed separately.
- Goal of this stage: reproducible feature detection and MS/MS-supported lipid annotation.
- This stage does not perform biological group statistics, pathway enrichment, or population-level inference.

2. Software / Files

The following software/files were actually used in the recorded workflow.

MS-DIAL Console Installation and Verification

The MS-DIAL Console package used in the tested workflow was MSDIAL.console.v5.5.251021-linux-net8.zip. The extracted executable was MSDIALCUI.

```
cd ~/Lipidomics
unzip MSDIAL.console.v5.5.251021-linux-net8.zip
chmod +x MSDIALCUI
./MSDIALCUI --help
```

The installed console returned:

Msdial Console App Version 5.5.251021

Msdial Console App requires the following args:

MsdialConsoleApp.exe <analysisType> -i <input folder> -o <output folder> -m <method file> -p (option)

<analysisType> is one of gcms, lcms

<input folder> is the folder containing the files to be processed

<output folder> is the folder to save results

<method file> is a file holding processing properties

-p is an option to generate MTB file to be loaded in MSDIAL GUI application.

Version verification: The installed executable, not the current MS-DIAL website release, is the version documented here: 5.5.251021.

Spectral Library Setup

Three library files were present in the MS-DIAL working directory. The MSP files were used for the final validated workflow. An LBM2 library was also tested during workflow development but was not retained as the final library route because the LBM2 loading step produced a MessagePack format error.

Library	Observed size	Workflow status
MSDIAL-InsilicoMSMS-Lipids-Pos.msp	70 MB	Used for positive-mode MSP annotation
MSDIAL-InsilicoMSMS-Lipids-Neg.msp	177 MB	Used for negative-mode MSP annotation
LipidMsmsBinaryDB-VS68-FiehnO.lbm2	121 MB	Tested; MessagePack loading error; not final

Library presence can be checked with:

```
ls -lh ~/Lipidomics/*.msp ~/Lipidomics/*.lbm2
```

The final parameter files point to the polarity-specific MSP libraries:

```
/home/shobita/Lipidomics/MSDIAL-InsilicoMSMS-Lipids-Pos.msp
/home/shobita/Lipidomics/MSDIAL-InsilicoMSMS-Lipids-Neg.msp
```

Item	Path
MS-DIAL Console 5.5.251021	~/Lipidomics/MSDIALCUI
Positive MSP library	~/Lipidomics/MSDIAL-InsilicoMSMS-Lipids-Pos.msp
Negative MSP library	~/Lipidomics/MSDIAL-InsilicoMSMS-Lipids-Neg.msp
Positive parameter file	~/Lipidomics/project/parameters/msdial5_lipidomics_POS_MSP.txt
Negative parameter file	~/Lipidomics/project/parameters/msdial5_lipidomics_NEG_MSP.txt
Python / pandas	Used only for reproducible extraction, annotation-code labeling, and QC summaries.

Why this check is done: to confirm the executable is present and to document the exact MS-DIAL Console version/syntax used for reproducibility.

3. Project Layout

```
~/Lipidomics/project/
├── data/
│   ├── mzml_pos/
│   └── mzml_neg/
├── parameters/
│   ├── msdial5_lipidomics_POS_MSP.txt
│   └── msdial5_lipidomics_NEG_MSP.txt
├── libraries/
├── results_pos_msp/
├── results_neg_msp/
├── results_summary/
├── scripts/
└── docs/
```

Why this structure is used: keeping POS and NEG inputs, parameters, and outputs in separate directories prevents accidental mixing of ion modes and makes the workflow easier to reproduce or later wrap in Nextflow.

4. mzML Verification

Purpose of mzML QC

These checks confirm that the expected mzML file exists, contains the expected number of spectra, contains MS1 and MS2 levels, and is centroided. This is an input-integrity check before MS-DIAL processing; it is not a biological QC test.

POSITIVE MODE

```
ls -lh ./data/mzml_pos/
grep -m1 -A1 "spectrumList count" ./data/mzml_pos/VZ0000099_PL_Pos.mzML
grep "<spectrum " ./data/mzml_pos/VZ0000099_PL_Pos.mzML | wc -l
grep 'name="ms level"' ./data/mzml_pos/VZ0000099_PL_Pos.mzML | head
grep 'value="2"' ./data/mzml_pos/VZ0000099_PL_Pos.mzML | wc -l
grep -i "centroid spectrum" ./data/mzml_pos/VZ0000099_PL_Pos.mzML | head
grep -i "profile spectrum" ./data/mzml_pos/VZ0000099_PL_Pos.mzML | head
```

NEGATIVE MODE

```
ls -lh ./data/mzml_neg/
grep -m1 -A1 "spectrumList count" ./data/mzml_neg/VZ0000099_PL_Neg.mzML
grep "<spectrum " ./data/mzml_neg/VZ0000099_PL_Neg.mzML | wc -l
grep 'name="ms level"' ./data/mzml_neg/VZ0000099_PL_Neg.mzML | head
grep 'value="2"' ./data/mzml_neg/VZ0000099_PL_Neg.mzML | wc -l
grep -i "centroid spectrum" ./data/mzml_neg/VZ0000099_PL_Neg.mzML | head
grep -i "profile spectrum" ./data/mzml_neg/VZ0000099_PL_Neg.mzML | head
```

RECORDED NEG RESULT the mzML contained 5,528 spectra, and MS1/MS2 entries and centroid-spectrum CV terms were observed in the XML inspection. The command counting value="2" also returned 7,433 because that search is not restricted to the MS-level CV parameter; it should therefore be treated as a broad XML text count, not an

MS2-spectrum count. This illustrates why the individual checks are diagnostic rather than a single pass/fail metric.

5. Parameter / Library Verification

Why this check is done: MS-DIAL annotation results depend directly on the parameter file and spectral library. Verifying file existence, size, and the library path recorded inside the parameter file helps ensure the run used the intended ion-mode-specific configuration.

POSITIVE MODE

```
ls -lh ./parameters/msdial5_lipidomics_POS_MSP.txt
ls -lh ~/Lipidomics/MSDIAL-InsilicoMSMS-Lipids-Pos.msp
grep -n "MSDIAL-InsilicoMSMS-Lipids-Pos" ./parameters/msdial5_lipidomics_POS_MSP.txt
```

NEGATIVE MODE

```
ls -lh ./parameters/msdial5_lipidomics_NEG_MSP.txt
ls -lh ~/Lipidomics/MSDIAL-InsilicoMSMS-Lipids-Neg.msp
grep -n "MSDIAL-InsilicoMSMS-Lipids-Neg" ./parameters/msdial5_lipidomics_NEG_MSP.txt
```

RECORDED NEG RESULT parameter file 12K; negative MSP library 177M; parameter file line 12 points to /home/shobita/Lipidomics/MSDIAL-InsilicoMSMS-Lipids-Neg.msp.

6. Recorded Parameters

Why parameter recording is necessary: a result cannot be reproduced from the output table alone. The important acquisition/processing settings must remain associated with the run.

POSITIVE MODE

Parameter	Value
General	Data type: Centroid • Ion mode: Positive • Target omics: Lipidomics • Ionization: ESI • Analysis type: LCMS
MSP library	MSDIAL-InsilicoMSMS-Lipids-Pos.msp
Adducts	[M+H] ⁺ , [M+NH ₄] ⁺ , [M+Na] ⁺ , [M+H-H ₂ O] ⁺
Tolerances	MS1: 0.01 • MS2: 0.025
Basic identification cutoff	80
Peak detection	LinearWeightedMovingAverage • smoothing 3 • minimum peak height 300 • minimum peak width 5 • average peak width 30
Deconvolution	Sigma 0.5 • max peaks 12 • exclude after precursor enabled
MSP annotation	Minimum spectrum match 3 • weighted/simple dot threshold 0.6 • reverse dot threshold 0.8 • total score 0.8 • MS1 0.01 • MS2 0.025

Parameter	Value
Alignment	RT tolerance 0.1 • spectrum similarity tolerance 0.8 • MS1 tolerance 0.015 • gap insertion enabled
Processing	All files • 20 threads

NEGATIVE MODE

Parameter	Value
General	Data type: Centroid • Ion mode: Negative • Target omics: Lipidomics • Ionization: ESI • Analysis type: LCMS
MSP library	MSDIAL-InsilicoMSMS-Lipids-Neg.msp
Adducts configured	[M-H] ⁻ and [M+HCOO] ⁻
Tolerances	MS1: 0.01 • MS2: 0.025
Basic identification cutoff	80
Peak detection / deconvolution / alignment / export	Retained as recorded in the negative parameter file, with the negative-ion lipid class/adduct configuration
MSP annotation	Minimum spectrum match 3 • weighted/simple dot threshold 0.6 • reverse dot threshold 0.8 • total score 0.8 • MS1 0.01 • MS2 0.025
Processing	All files • 20 threads

Mobile-phase/additive context recorded for the experiment: 10 mM ammonium formate + 0.1% formic acid. The negative-mode experiment additionally used ammonium formate as the recorded solvent/additive context.

7. MS-DIAL Command

Why this step is done: this is the primary computational processing step. MS-DIAL performs LC-MS peak detection, deconvolution, spectral annotation, and output generation according to the supplied parameter file.

POSITIVE MODE

```
cd ~/Lipidomics/project
mkdir -p results_pos_msp
~/Lipidomics/MSDIALCUI lcms \
-i ./data/mzml_pos/ \
-o ./results_pos_msp/ \
-m ./parameters/msdial5_lipidomics_POS_MSP.txt \
-p
```

NEGATIVE MODE

```
cd ~/Lipidomics/project
mkdir -p results_neg_msp
```

```
~/Lipidomics/MSDIALCUI lcms \  
-i ./data/mzml_neg/ \  
-o ./results_neg_msp/ \  
-m ./parameters/msdial5_lipidomics_NEG_MSP.txt \  
-p
```

The console should report stages such as Loading analysis files, Start processing, Reading data, Peak picking started, Deconvolution started, and Annotation started.

8. Output Verification

Why this check is done: a successful process message is not sufficient. The expected MS-DIAL output files should be present so that the run can be traced and downstream tables can be generated.

POSITIVE MODE

```
ls -lh results_pos_msp/  
find results_pos_msp -maxdepth 1 -type f -printf "%f\t%s bytes\n" | sort
```

NEGATIVE MODE

```
ls -lh results_neg_msp/  
find results_neg_msp -maxdepth 1 -type f -printf "%f\t%s bytes\n" | sort
```

RECORDED NEG OUTPUTS AlignResult-2026911119.mdalign, AlignResult-2026911119.mdmsp, AlignResult-2026911119.mzTabM, Project-2609111109.mdproject, VZ0000099_PL_Neg.mdmsp, and VZ0000099_PL_Neg.mdpeak.

The .mdpeak file is the principal feature-level table for this workflow because it contains peak information, annotation fields, matching metrics, adduct information, and MS-DIAL annotation tags. The .mdmsp and alignment/project files are retained for traceability.

EXECUTION NOTE Run **ONLY** the **POSITIVE MODE** subsection for a positive-ion dataset and **ONLY** the **NEGATIVE MODE** subsection for a negative-ion dataset. The two commands use different input folders, output folders, and parameter files, so they are intentionally independent.

9. Extract MS/MS Annotations

Why this check/extraction is done: the complete .mdpeak contains all detected features, including features without MS/MS-supported annotation. For downstream lipid annotation, a separate table containing only MS/MS-matched features provides a focused, reproducible hand-off while preserving the complete .mdpeak as the source record.

```
cd ~/Lipidomics/project
```

POSITIVE MODE

```
SAMPLE="VZ0000099_PL_Pos"  
MDPEAK="results_pos_msp/${SAMPLE}.mdpeak"
```

```
OUT="results_summary/${SAMPLE}_FINAL_MSMS_annotations.tsv"
export SAMPLE MDPEAK OUT

python - <<'PY'
import os
import pandas as pd

sample=os.environ["SAMPLE"]
mdpeak=os.environ["MDPEAK"]
out=os.environ["OUT"]

df=pd.read_csv(mdpeak,sep="\t",low_memory=False)

def is_true(x):
    return str(x).strip().lower() in {"true","1","yes"}

if "MS/MS matched" not in df.columns:
    raise SystemExit("ERROR: 'MS/MS matched' column not found in .mdpeak")

msms=df[df["MS/MS matched"].map(is_true)].copy()

wanted=[
    "Peak ID","Name","Ontology","Adduct","RT (min)","Precursor m/z",
    "Height","Area","Formula","Reference m/z","m/z similarity",
    "Simple dot product","Weighted dot product","Reverse dot product",
    "Matched peaks count","Matched peaks percentage","Total score","S/N",
    "Annotation tag (VS1.0)"
]

missing=[c for c in wanted if c not in msms.columns]
if missing:
    raise SystemExit("ERROR: Missing columns: "+", ".join(missing))

msms=msms[wanted].rename(columns={
    "Annotation tag (VS1.0)": "MS-DIAL_annotation_code"
})
msms.to_csv(out,sep="\t",index=False)

print("Total .mdpeak features:",len(df))
print("MS/MS-matched features:",len(msms))
print("Saved:",out)
PY
```

NEGATIVE MODE

```
SAMPLE="VZ0000099_PL_Neg"
MDPEAK="results_neg_msp/${SAMPLE}.mdpeak"
OUT="results_summary/${SAMPLE}_FINAL_MSMS_annotations.tsv"
export SAMPLE MDPEAK OUT

python - <<'PY'
```

```

import os
import pandas as pd

mdpeak=os.environ["MDPEAK"]
out=os.environ["OUT"]

df=pd.read_csv(mdpeak,sep="\t",low_memory=False)

def is_true(x):
    return str(x).strip().lower() in {"true","1","yes"}

msms=df[df["MS/MS matched"].map(is_true)].copy()

wanted=[
    "Peak ID","Name","Ontology","Adduct","RT (min)","Precursor m/z",
    "Height","Area","Formula","Reference m/z","m/z similarity",
    "Simple dot product","Weighted dot product","Reverse dot product",
    "Matched peaks count","Matched peaks percentage","Total score","S/N",
    "Annotation tag (VS1.0)"
]

msms=msms[wanted].rename(columns={
    "Annotation tag (VS1.0)": "MS-DIAL_annotation_code"
})
msms.to_csv(out,sep="\t",index=False)

print("Total .mdpeak features:",len(df))
print("MS/MS-matched features:",len(msms))
print("Saved:",out)
PY

```

RECORDED RESULT 13,032 total features and 472 MS/MS-matched features (POS) • 16,036 total features and 248 MS/MS-matched features (NEG).

EXECUTION NOTE Run the POSITIVE MODE subsection on the POSITIVE_MODE FINAL_MSMS_annotations.tsv file and the NEGATIVE MODE subsection on the NEGATIVE_MODE FINAL_MSMS_annotations.tsv file. Do not substitute one mode's input/output filenames into the other mode's command.

10. Correct Annotation Levels

Why this step is done: the numeric MS-DIAL annotation tag needs a human-readable label before the table is used downstream. The mapping is kept explicit so that annotation levels are not misinterpreted.

For this lipidomics handover, code 410 is labelled 'Lipid acyl-chain level' and code 420 'Lipid class level'. Code 430 is deliberately labelled 'MS-DIAL code 430 (not a lipidomics-specific resolved level)' rather than being called an acyl-position level.

POSITIVE MODE

```
SAMPLE="VZ0000099_PL_Pos"
IN="results_summary/${SAMPLE}_FINAL_MSMS_annotations.tsv"
OUT="results_summary/${SAMPLE}_FINAL_MSMS_annotations_CORRECTED.tsv"
export SAMPLE IN OUT

python - <<'PY'
import os
import pandas as pd

inp=os.environ["IN"]
out=os.environ["OUT"]
df=pd.read_csv(inp,sep="\t",low_memory=False)

def map_annotation(code):
    try: code=int(code)
    except: return "Unknown / unclassified"
    if code==410: return "Lipid acyl-chain level"
    if code==420: return "Lipid class level"
    if code==430: return "MS-DIAL code 430 (not a lipidomics-specific resolved level)"
    if code==999: return "Unknown"
    return f"MS-DIAL code {code}"

df["MS-DIAL_annotation_level"]=df["MS-DIAL_annotation_code"].apply(map_annotation)
df.to_csv(out,sep="\t",index=False)

print(df["MS-DIAL_annotation_level"].value_counts().to_string())
print("\nAnnotation code distribution:")
print(df["MS-DIAL_annotation_code"].value_counts().sort_index().to_string())
print("\nSaved:",out)
PY
```

NEGATIVE MODE

```
SAMPLE="VZ0000099_PL_Neg"
IN="results_summary/${SAMPLE}_FINAL_MSMS_annotations.tsv"
OUT="results_summary/${SAMPLE}_FINAL_MSMS_annotations_CORRECTED.tsv"
export SAMPLE IN OUT

python - <<'PY'
import os
import pandas as pd

inp=os.environ["IN"]
out=os.environ["OUT"]
df=pd.read_csv(inp,sep="\t",low_memory=False)

def map_annotation(code):
    try: code=int(code)
    except: return "Unknown / unclassified"
    if code==410: return "Lipid acyl-chain level"
    if code==420: return "Lipid class level"
    if code==430: return "MS-DIAL code 430 (not a lipidomics-specific resolved level)"
```

```

    if code==999: return "Unknown"
    return f"MS-DIAL code {code}"

df["MS-DIAL_annotation_level"]=df["MS-
DIAL_annotation_code"].apply(map_annotation)
df.to_csv(out,sep="\t",index=False)

print(df["MS-DIAL_annotation_level"].value_counts().to_string())
print("\nAnnotation code distribution:")
print(df["MS-
DIAL_annotation_code"].value_counts().sort_index().to_string())
print("\nSaved:",out)
PY

```

Annotation code	Label	POS count	NEG count
410	Lipid acyl-chain level	131	122
420	Lipid class level	184	102
430	MS-DIAL code 430 (not a lipidomics-specific resolved level)	157	24

11. MS/MS QC Summary

Why these QC checks are done: MS/MS annotation is evidence-based, but not every MS/MS match has the same amount or quality of fragment evidence. The following summary describes the observed annotation evidence and is intended for review and reporting—not for inventing new biological thresholds or automatically deleting features.

- Total detected features: measures the number of chromatographic features generated by peak picking.
- m/z-matched features: indicates features for which MS-DIAL found a library/theoretical mass correspondence under the configured matching criteria.
- MS/MS-matched features: counts features with MS/MS spectral matching; this is the primary evidence subset used for the MS/MS annotation table.
- MS/MS annotation table rows: verifies that the extracted table contains the expected number of MS/MS-supported rows.
- Adduct distribution: checks which ion/adduct forms are represented and helps interpret precursor measurements.
- MS-DIAL annotation-code distribution: documents the annotation categories assigned by MS-DIAL.
- Matched peaks count: describes how many experimental fragment peaks contributed to each library-spectrum match; higher counts generally indicate more fragment-level evidence, but the count alone is not a validation threshold.

QC summary command — POSITIVE

```
cd ~/Lipidomics/project
SAMPLE="VZ0000099_PL_Pos"
MDPEAK="results_pos_msp/${SAMPLE}.mdpeak"
ANN="results_summary/${SAMPLE}_FINAL_MSMS_annotations_CORRECTED.tsv"
QC="results_summary/${SAMPLE}_QC_summary.tsv"
export SAMPLE MDPEAK ANN QC

python - <<'PY'
import os
import pandas as pd

df=pd.read_csv(os.environ["MDPEAK"],sep="\t",low_memory=False)
ann_df=pd.read_csv(os.environ["ANN"],sep="\t",low_memory=False)

def count_true(s):
    return s.astype(str).str.strip().str.lower().isin({"true","1","yes"}).sum()

rows=[
("Total detected features",len(df)),
("m/z-matched features",count_true(df["m/z matched"])),
("MS/MS-matched features",count_true(df["MS/MS matched"])),
("MS/MS annotation table rows",len(ann_df))
]

for k,v in ann_df["Adduct"].value_counts(dropna=False).items():
    rows.append((f"MS/MS annotations – Adduct: {k}",int(v)))

for k,v in ann_df["MS-DIAL_annotation_code"].value_counts(dropna=False).sort_index().items():
    rows.append((f"MS-DIAL annotation code: {k}",int(v)))

for k,v in ann_df["Matched peaks count"].value_counts(dropna=False).sort_index().items():
    rows.append((f"Matched peaks count: {k}",int(v)))

out=pd.DataFrame(rows,columns=["Metric","Value"])
out.to_csv(os.environ["QC"],sep="\t",index=False)
print(out.to_string(index=False))
print("\nSaved:",os.environ["QC"])
PY
```

QC summary command — NEGATIVE

```
cd ~/Lipidomics/project
SAMPLE="VZ0000099_PL_Neg"
MDPEAK="results_neg_msp/${SAMPLE}.mdpeak"
ANN="results_summary/${SAMPLE}_FINAL_MSMS_annotations_CORRECTED.tsv"
QC="results_summary/${SAMPLE}_QC_summary.tsv"
export SAMPLE MDPEAK ANN QC
```

```
python - <<'PY'
import os
import pandas as pd

df=pd.read_csv(os.environ["MDPEAK"],sep="\t",low_memory=False)
ann_df=pd.read_csv(os.environ["ANN"],sep="\t",low_memory=False)

def count_true(s):
    return
s.astype(str).str.strip().str.lower().isin({"true","1","yes"}).sum()

rows=[
("Total detected features",len(df)),
("m/z-matched features",count_true(df["m/z matched"])),
("MS/MS-matched features",count_true(df["MS/MS matched"])),
("MS/MS annotation table rows",len(ann_df))
]

for k,v in ann_df["Adduct"].value_counts(dropna=False).items():
    rows.append((f"MS/MS annotations – Adduct: {k}",int(v)))

for k,v in ann_df["MS-
DIAL_annotation_code"].value_counts(dropna=False).sort_index().items():
    rows.append((f"MS-DIAL annotation code: {k}",int(v)))

for k,v in ann_df["Matched peaks
count"].value_counts(dropna=False).sort_index().items():
    rows.append((f"Matched peaks count: {k}",int(v)))

out=pd.DataFrame(rows,columns=["Metric","Value"])
out.to_csv(os.environ["QC"],sep="\t",index=False)
print(out.to_string(index=False))
print("\nSaved:",os.environ["QC"])
PY
```

Metric	NEG value
Detected features	16,036
m/z-matched features	3,254
MS/MS-matched features	248
Annotation-table rows	248
Adduct: [M+HCOO]–	189
Adduct: [M-H]–	57
Adduct: [M+CH3COO]–	2
Annotation code 410	122
Annotation code 420	102
Annotation code 430	24

Matched-peaks count 1 / 2 / 3 / 4 / 5	8 / 15 / 73 / 42 / 21
Matched-peaks count 6 / 7 / 8 / 9 / 12	10 / 11 / 52 / 13 / 3

Recorded NEG QC summary — full metric breakdown from the run.

12. Handling Unannotated Features

Why this distinction is important: MS-DIAL detects many chromatographic features that do not obtain an MS/MS-supported library annotation. These features are real computational detections, but they should not automatically be treated as identified lipids.

- Retain the complete .mdpeak file for traceability and future re-annotation.
- Use the MS/MS-matched table as the primary annotated-lipid hand-off from this stage.
- Do not assign a lipid name solely because a feature has an m/z value.
- Do not discard unannotated features merely because they are unannotated; they remain available for future library or method development.
- Do not convert an MS-DIAL annotation code into a stronger structural claim than the documented code supports.

13. Output Hand-off

Why this step is done: it defines exactly which files should be carried into downstream analysis so that later analysts do not have to infer which MS-DIAL output was intended.

9A. POSITIVE MODE — Extract MS/MS Annotations

File	Path
Primary source	results_pos_msp/VZ0000099_PL_Pos.mdpeak
MS/MS annotation table	results_summary/VZ0000099_PL_Pos_FINAL_MSMS_annotations.tsv
Corrected annotation-level table	results_summary/VZ0000099_PL_Pos_FINAL_MSMS_annotations_CORRECTED.tsv
QC summary	results_summary/VZ0000099_PL_Pos_QC_summary.tsv

9B. NEGATIVE MODE — Extract MS/MS Annotations

File	Path
Primary source	results_neg_msp/VZ0000099_PL_Neg.mdpeak
MS/MS annotation table	results_summary/VZ0000099_PL_Neg_FINAL_MSMS_annotations.tsv
Corrected annotation-level table	results_summary/VZ0000099_PL_Neg_FINAL_MSMS_annotations_CORRECTED.tsv
QC summary	results_summary/VZ0000099_PL_Neg_QC_summary.tsv

Keep the POS and NEG tables separate until both mode-specific workflows are finalized. Only then should they be combined using an explicit, documented integration step.

14. Population / Downstream Hand-off

Why this section is included: MS-DIAL annotation is the feature-identification stage. Population-level statistics require a matrix of comparable samples, consistent feature definitions, and an explicit study design.

- For each sample and ion mode, carry forward the lipid annotation fields, precursor m/z, RT, area/height, adduct, annotation code/level, and MS/MS similarity metrics.
- Use full lipid names only after the structural annotation level is appropriate; do not manufacture full structures from class-level evidence.
- For POS and NEG, maintain separate mode-specific feature tables before integration.
- For population statistics, define the biological groups/phenotypes, replicate structure, missing-value handling, normalization, and statistical model before calculating P values or FDR.
- Pathway/functional analysis should be performed only after lipid identities have been mapped to an appropriate downstream identifier set. The present MS-DIAL stage does not itself establish pathway significance.
- If a single sample has no biological comparison group, P values and FDR for differential abundance are not meaningful at this stage.

15. Reproducibility

Why this is done: a handover should allow another analyst to reproduce the processing without relying on undocumented GUI settings or memory.

- Keep the exact MS-DIAL Console version.
- Keep the exact POS and NEG parameter files unchanged alongside the results.
- Keep the exact MSP library files or their controlled versions.
- Keep the original mzML files.
- Keep the command history used to generate each result.
- Do not overwrite POS results with NEG results or vice versa.
- Keep the complete .mdpeak outputs even after creating filtered/extracted tables.
- Record any future parameter/library changes as a new version rather than silently replacing the original configuration.

16. References

- MS-DIAL official documentation: annotation codes and levels — <https://systemsomicslab.github.io/compms/msdial/annotationcode.html>
- MS-DIAL official main documentation — <https://systemsomicslab.github.io/compms/msdial/main>
- MS-DIAL 5 lipidomics tutorial — <https://systemsomicslab.github.io/msdial5tutorial/lipidomics.html>
- MS-DIAL lipid nomenclature documentation — <https://systemsomicslab.github.io/compms/msdial/lipidnomenclature.html>

- MS-DIAL Console application documentation — <https://systemsomicslab.github.io/compms/msdial/consoleapp.html>

Appendix A — Recorded NEG Input/Output Evidence

The negative-mode run recorded in the handover produced 5,528 spectra in the mzML, completed MS-DIAL processing through peak picking, deconvolution and annotation, and generated the expected .mdpeak/.mdmsp/alignment/project outputs. The extracted MS/MS table contained 248 features from 16,036 detected features.

WARNINGS OBSERVED MS-DIAL emitted the warnings 'cvParam MS:1000522 is not supported yet' and 'cvParam (unit) UO:0000110 is not supported yet' during reading. The run continued through peak picking, deconvolution and annotation and produced output files. These messages should therefore be retained in the run record rather than silently omitted.

Appendix B — Recommended Interpretation Hierarchy

Output	What it represents	Use
*.mdpeak	All detected MS-DIAL features plus annotation/QC fields	Primary traceable source
FINAL_MSMS_annotations.tsv	MS/MS-matched features only	Focused annotation hand-off
FINAL_MSMS_annotations_CORRECTED.tsv	Same MS/MS table with explicit annotation-level labels	Preferred annotation hand-off
*_QC_summary.tsv	Run-level feature/annotation counts	QC and reporting
*.mdmsp / *.mdalign / *.mdproject	MS-DIAL spectral/alignment/project outputs	Reproducibility and traceability